Infrastructure as Code (IaC) - Project Documentation

Capstone Project 3: Core Cloud Resources with Terraform

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REGION
Asia Pacific (Mumbai) - ap-south-1

PROVIDER
AWS | Terraform v1.15.1

TOTAL RESOURCES
22 across 4 modules

DATE
May 6, 2026

REPOSITORY
rakeshduvva/capstone-capg-task3

12	2	5	3
NETWORKING	COMPUTE	STORAGE	DATABASE

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1. Architecture Overview

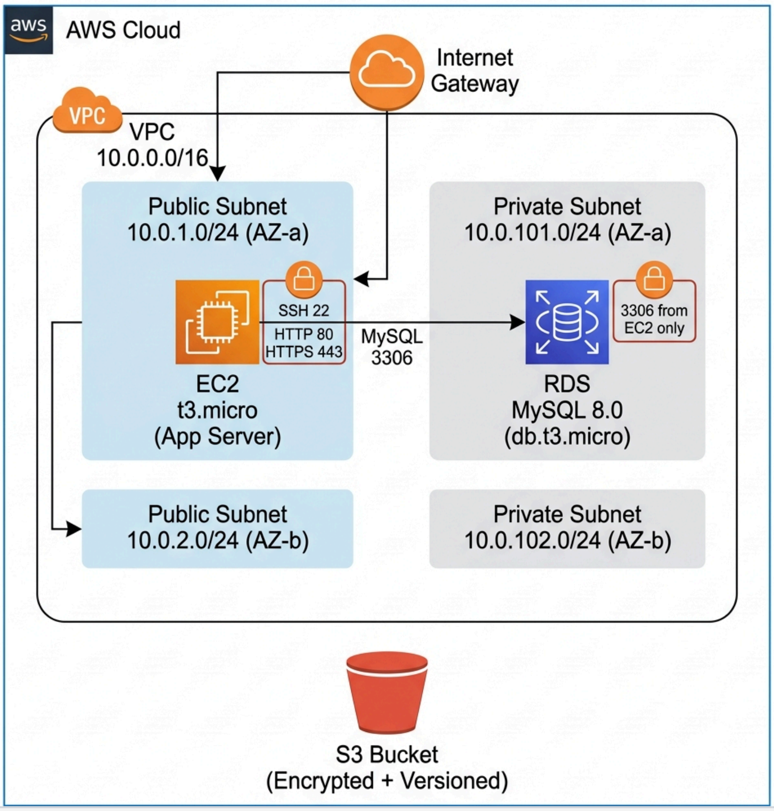

Figure 1.1: *Cloud architecture diagram showing VPC (10.0.0.0/16) with public/private subnets across two AZs, EC2 in public subnet, RDS in private subnet, Internet Gateway, and S3 bucket.*

Module	Resources	Description
Networking	12	VPC, 4 Subnets, IGW, 2 Route Tables, 4 Route Table Associations
Compute	2	EC2 Instance (t3.micro), Security Group
Storage	5	S3 Bucket, Versioning, Encryption, Lifecycle, Public Access Block
Database	3	RDS MySQL (db.t3.micro), DB Subnet Group, Security Group

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2. Terraform Workflow

2.1 Terraform Init

Initializes the working directory, downloads the AWS provider, and prepares the modules.

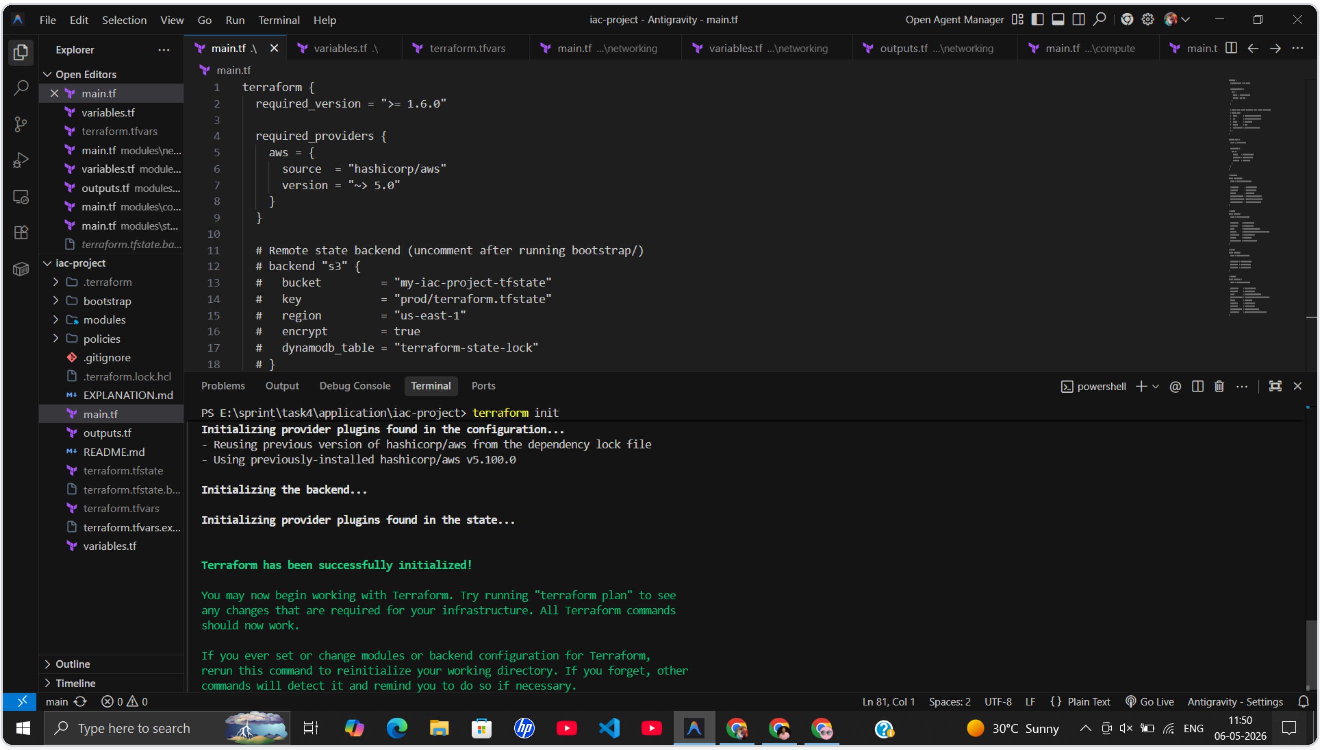

Figure 2.1: *`terraform init` completing successfully — hashicorp/aws v5.100.0 provider installed, backend initialized, provider plugins found in state.*

2.2 Terraform Plan

Previews the execution plan showing all 22 resources that will be created.

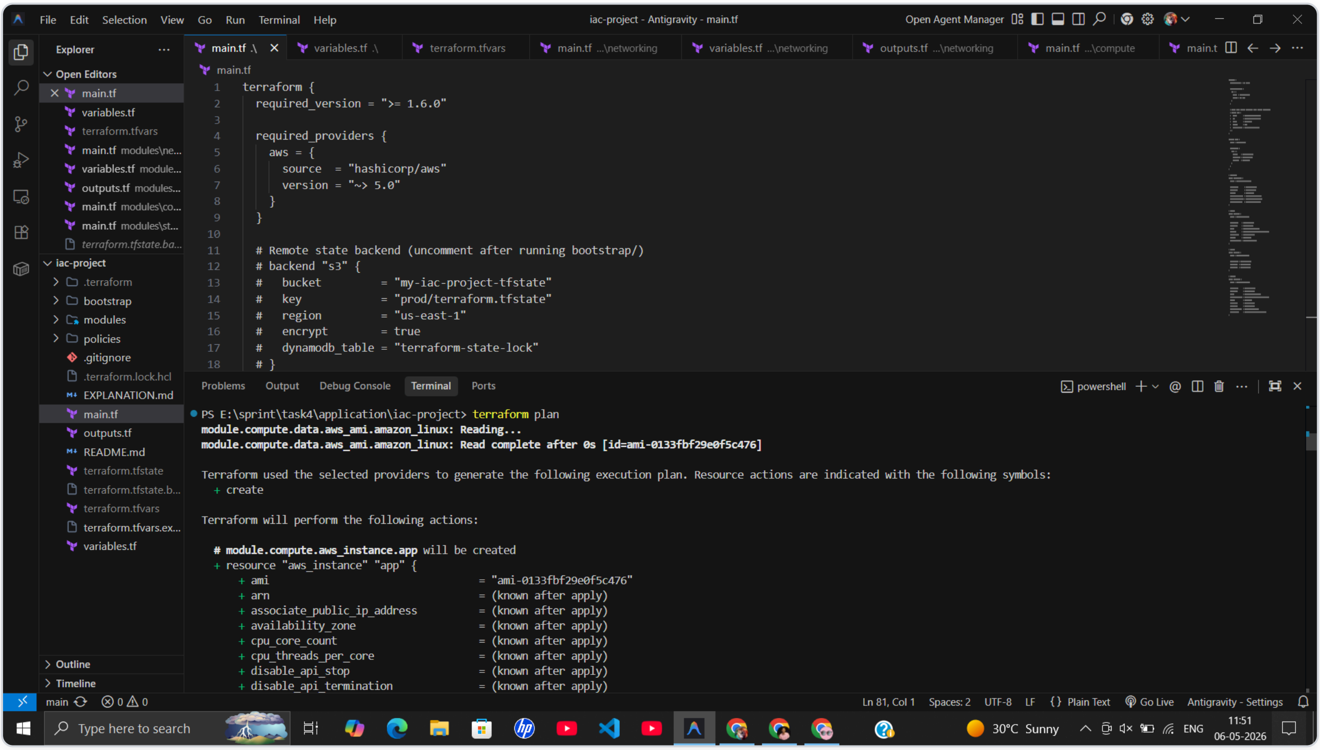

Figure 2.2: *`terraform plan` reads the Amazon Linux AMI data source and begins listing resources to create, starting with `module.compute.aws_instance.app`.*

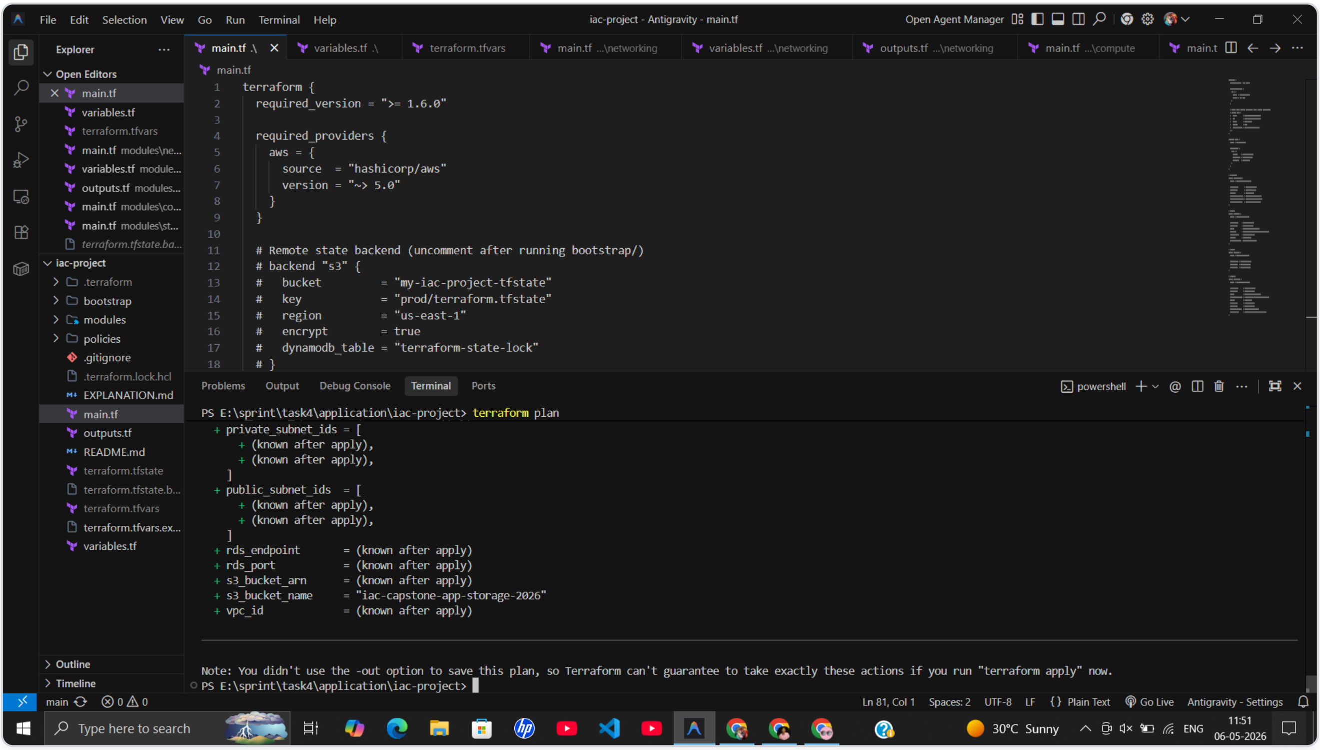

Figure 2.3: *Plan summary showing planned outputs (subnet IDs, RDS endpoint, S3 bucket details, VPC ID). Confirms:* **"22 to add, 0 to change, 0 to destroy"**.

2.3 Terraform Apply (First Deployment)

Deploys all 22 resources to AWS. The RDS instance takes ~7 minutes to provision.

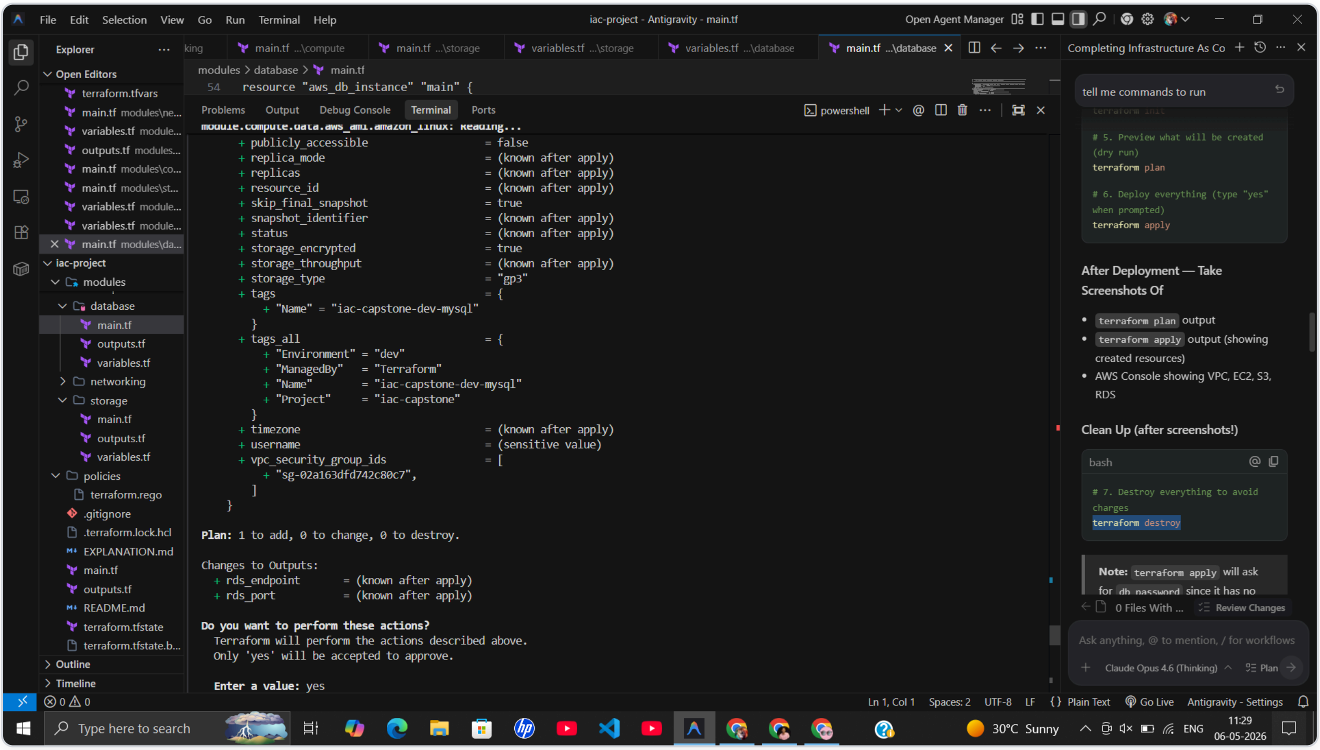

Figure 2.4: *`terraform apply` showing the database module plan with `aws_db_instance` resource details — encryption enabled, storage type gp3, skip_final_snapshot true.*

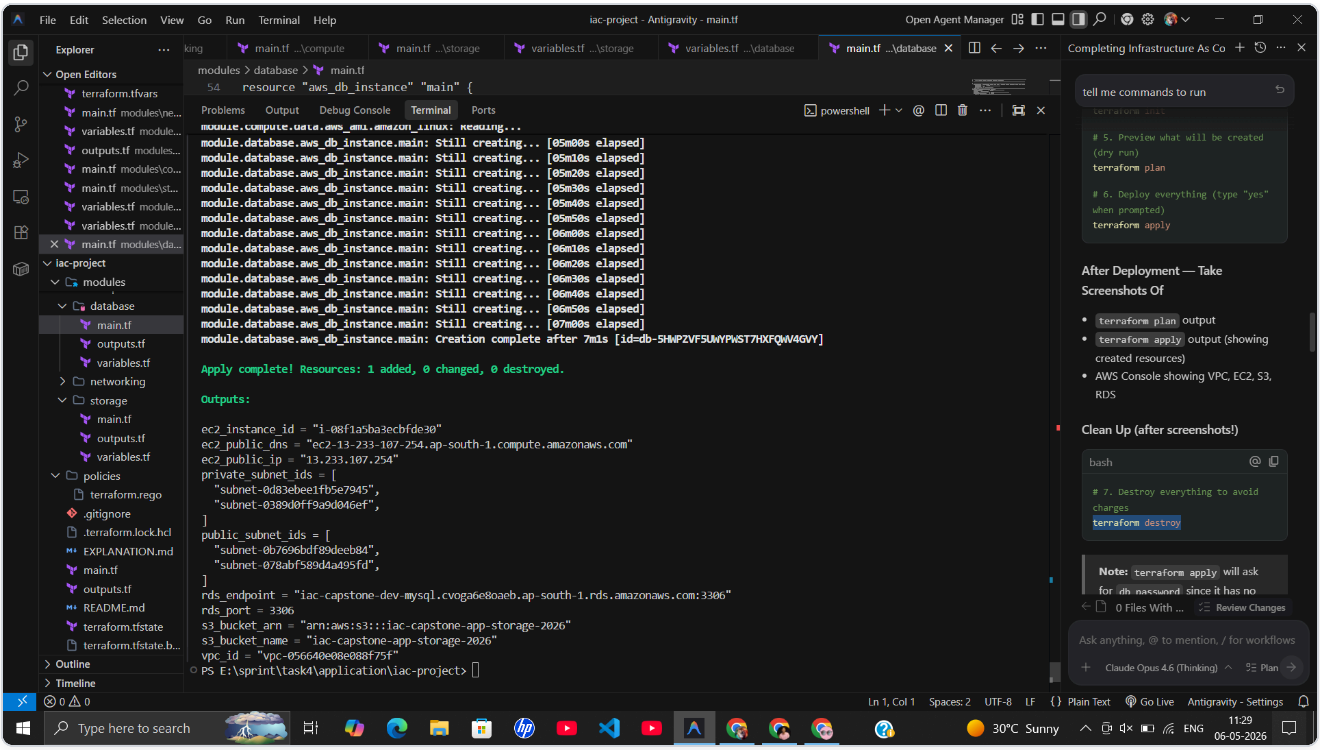

Figure 2.5: *Apply complete. RDS creation took ~7 minutes. Outputs: EC2 Instance ID* `i-08f1a5ba3ecbfde30` *, Public IP* `13.233.107.254` *, S3 ARN* `arn:aws:s3:::iac-capstone-app-storage-2026` *, VPC* `vpc-056640e08e088f75f` *.*

2.4 Terraform Destroy (First Teardown)

Tears down all 22 resources to avoid AWS charges.

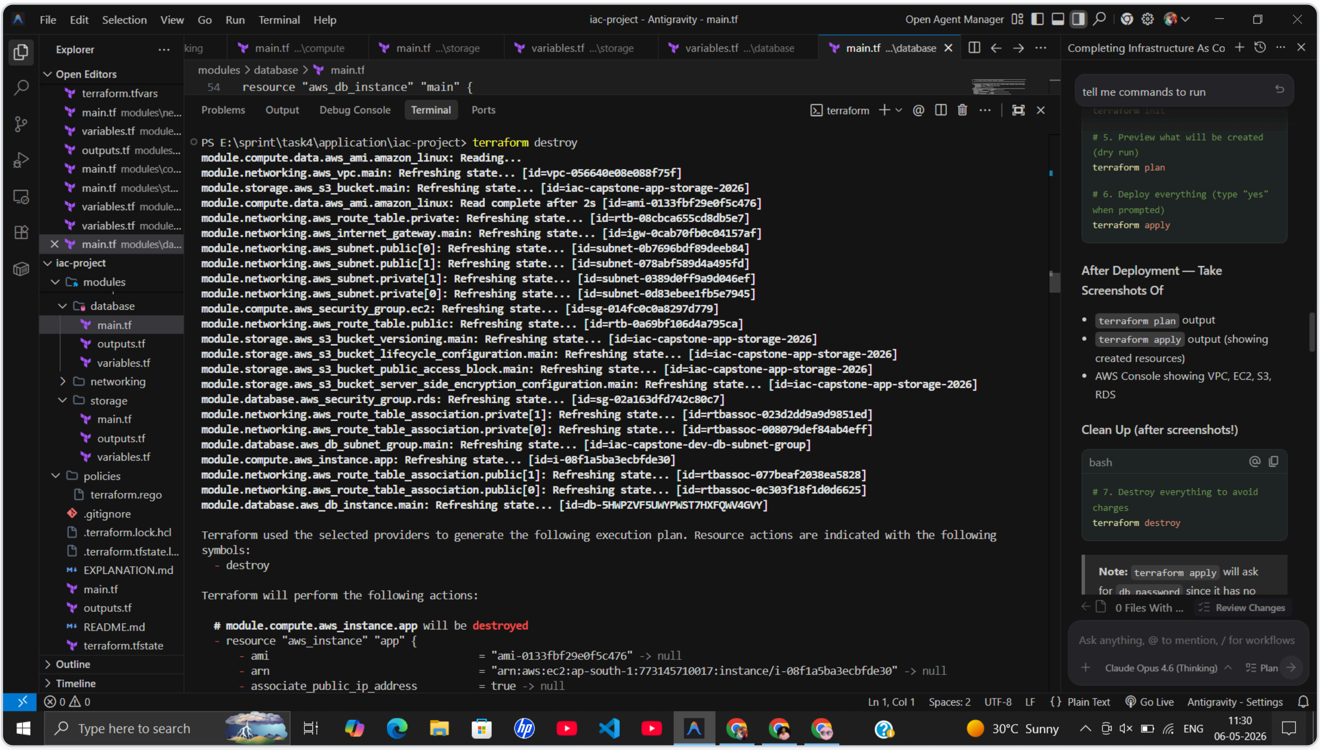

Figure 2.6: `terraform destroy` *refreshing state of all 22 resources before generating the destruction plan.*

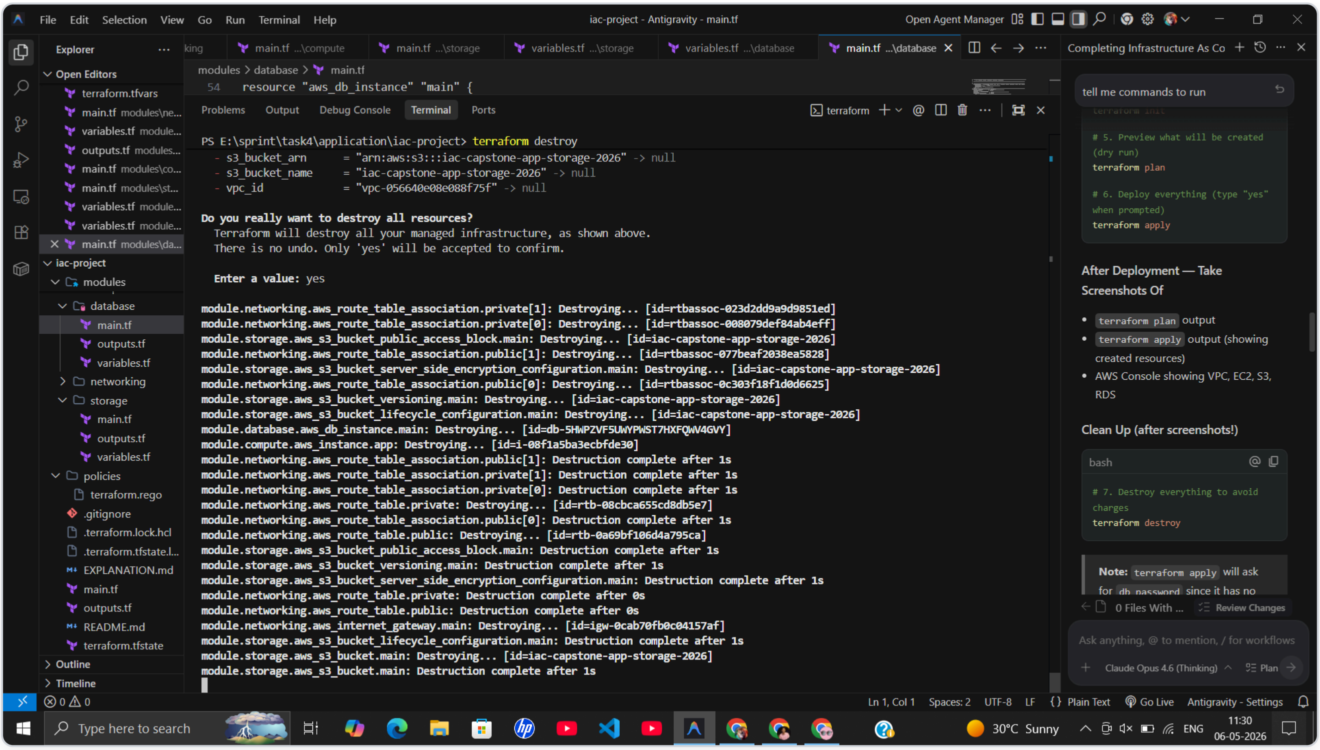

Figure 2.7: *Destruction plan showing detailed attributes of each resource to be removed, including subnet CIDR blocks, availability zones, and tags.*

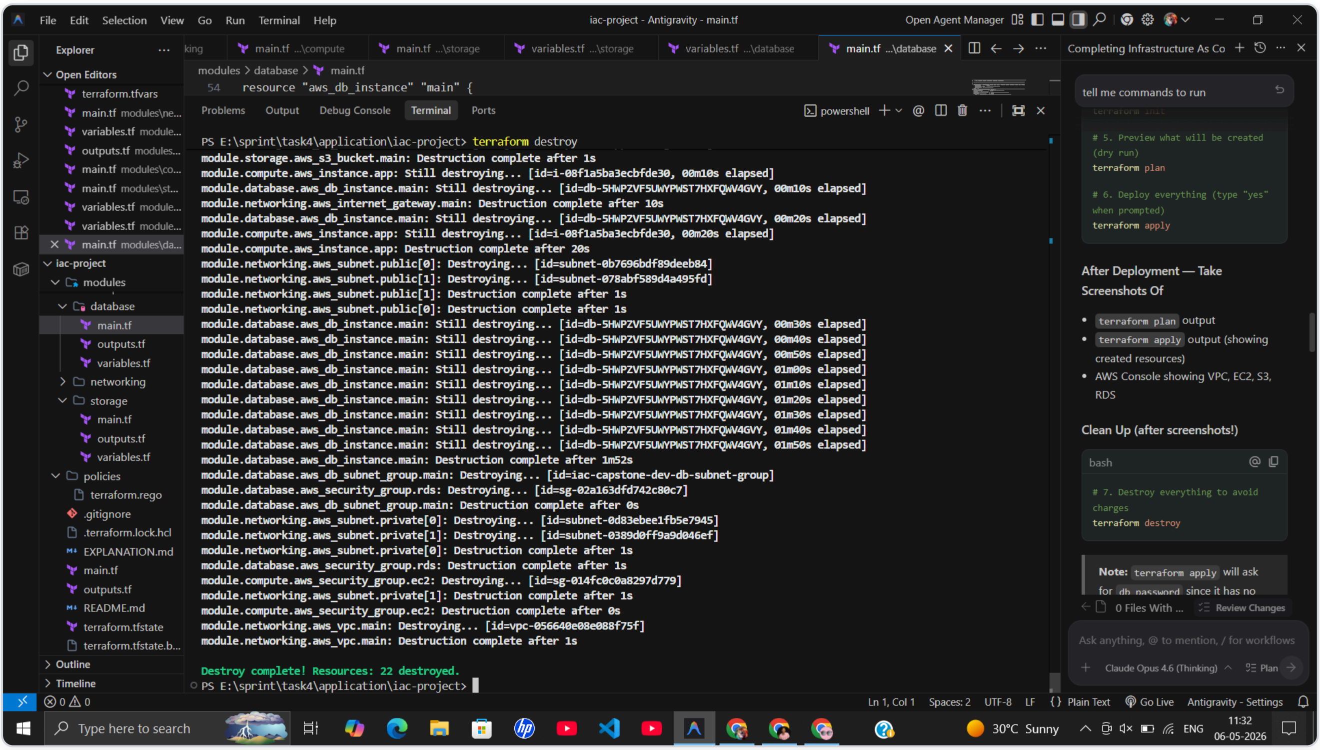

Figure 2.8: *Destroy in progress — resources removed in dependency order. Final output:* **"Destroy complete! Resources: 22 destroyed."**

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3. AWS Console Verification

The following screenshots verify that all Terraform-managed resources were successfully provisioned in the AWS Management Console.

3.1 Networking

VPC

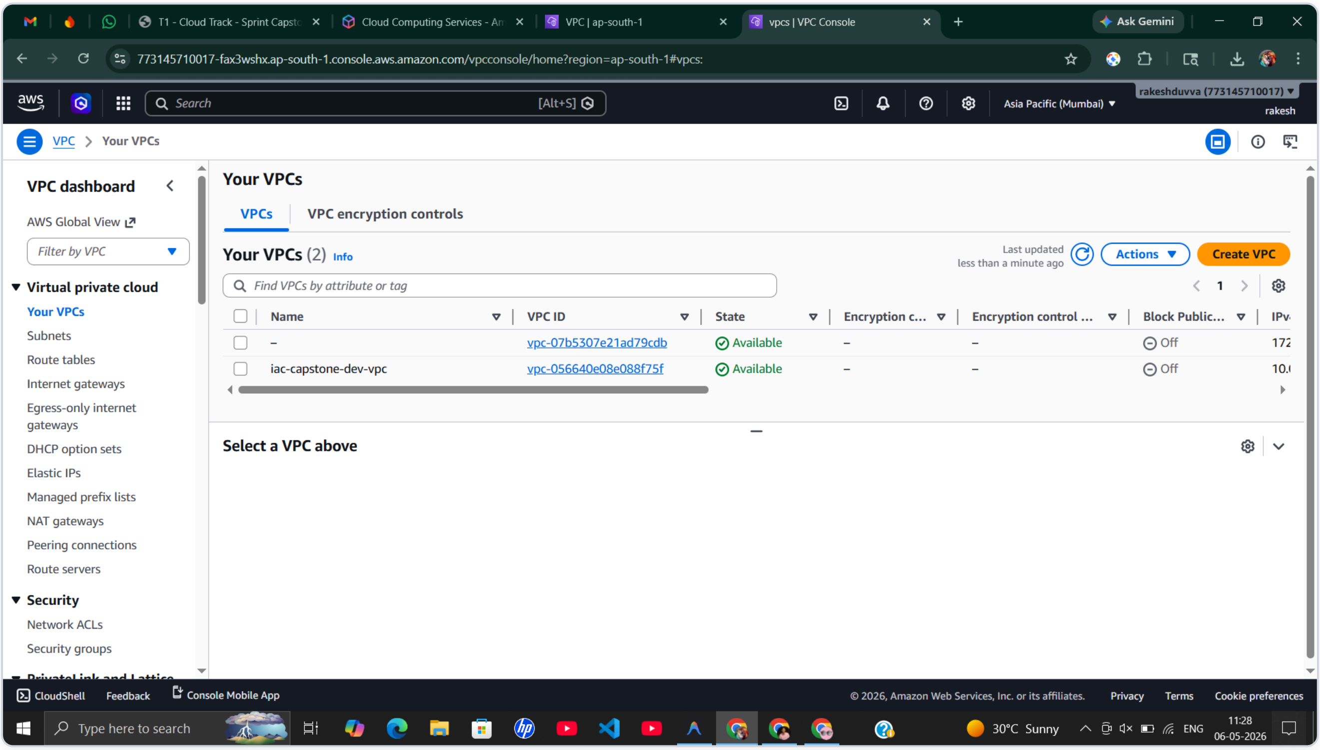

Figure 3.1: *VPC Console showing* `iac-capstone-dev-vpc` *(vpc-056640e08e088f75f) in "Available" state alongside the default VPC.*

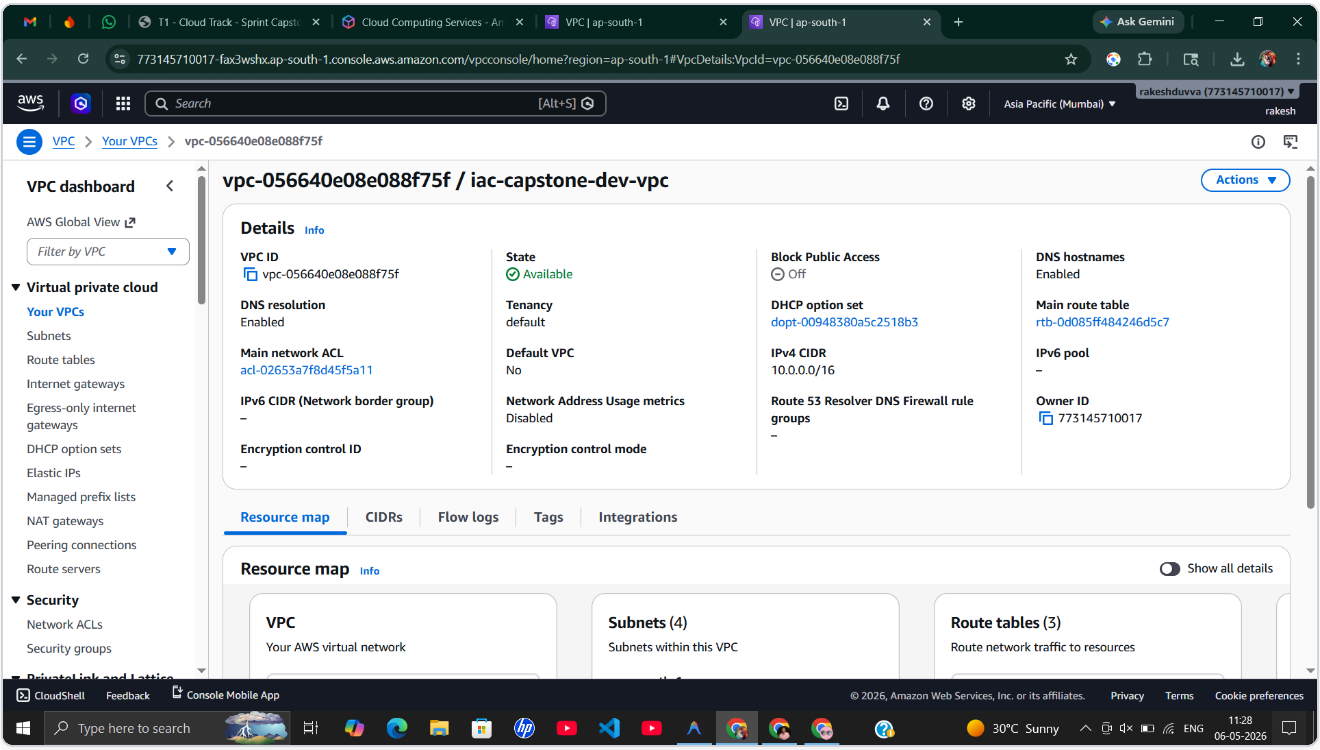

Figure 3.2: *VPC details — IPv4 CIDR:* `10.0.0.0/16` *, DNS resolution: Enabled, DNS hostnames: Enabled, Tenancy: default, Main route table and DHCP options set configured.*

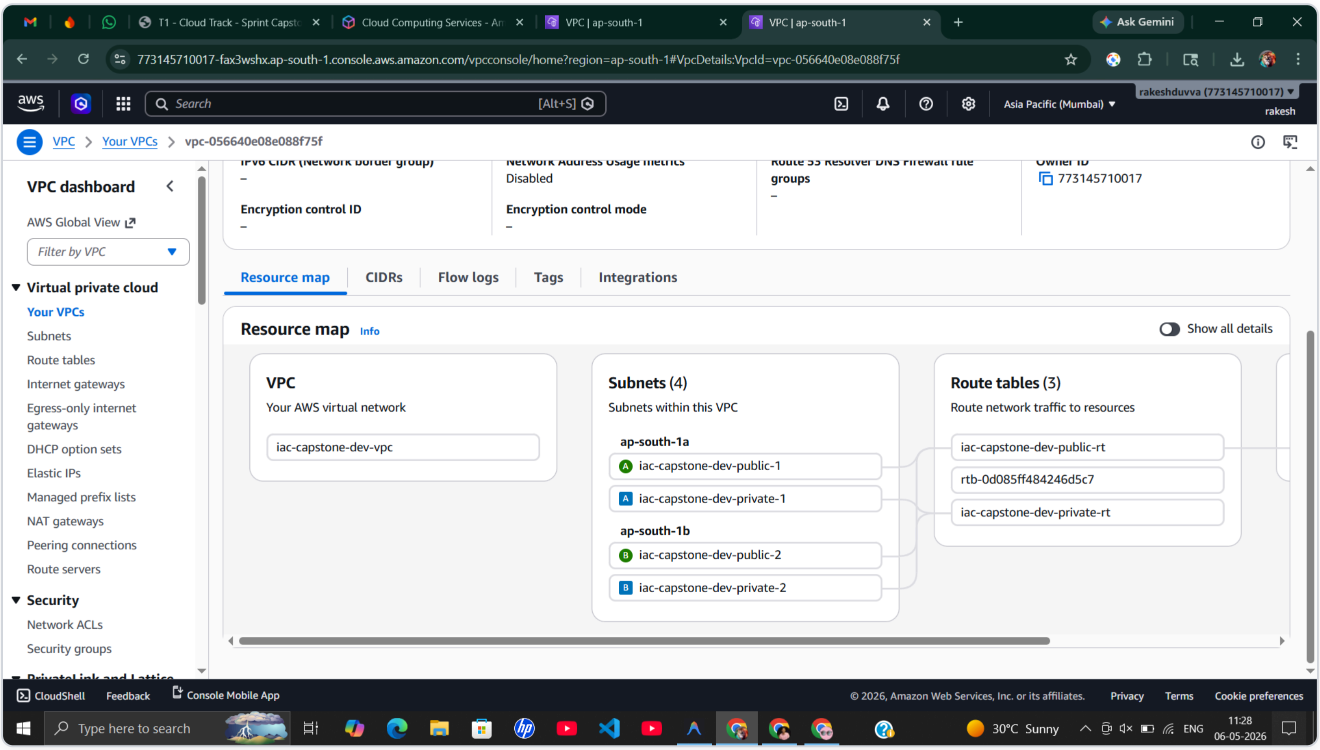

Figure 3.3: *VPC Resource Map showing the complete network topology: 1 VPC, 4 Subnets (2 public + 2 private across ap-south-1a and 1b), 3 Route Tables.*

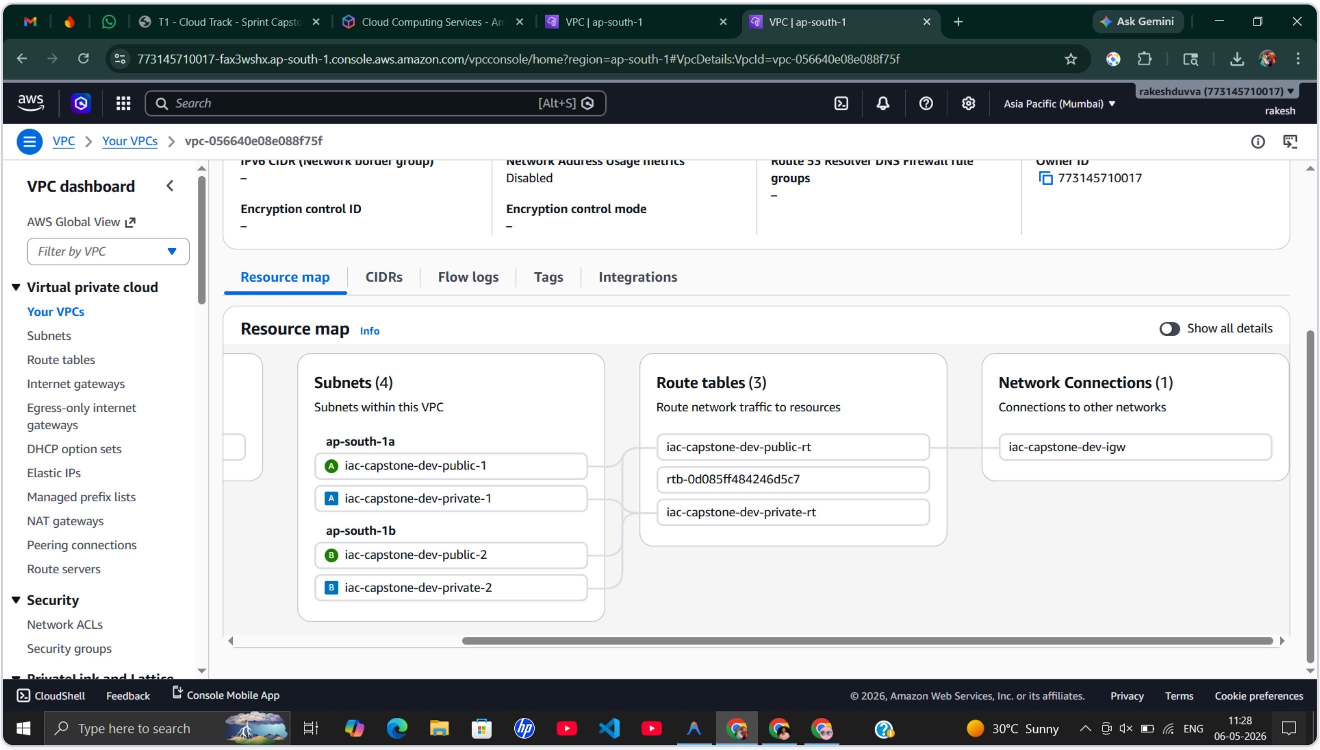

Figure 3.4: *Extended resource map showing Network Connections with the Internet Gateway* `iac-capstone-dev-igw` *connected to the public route table.*

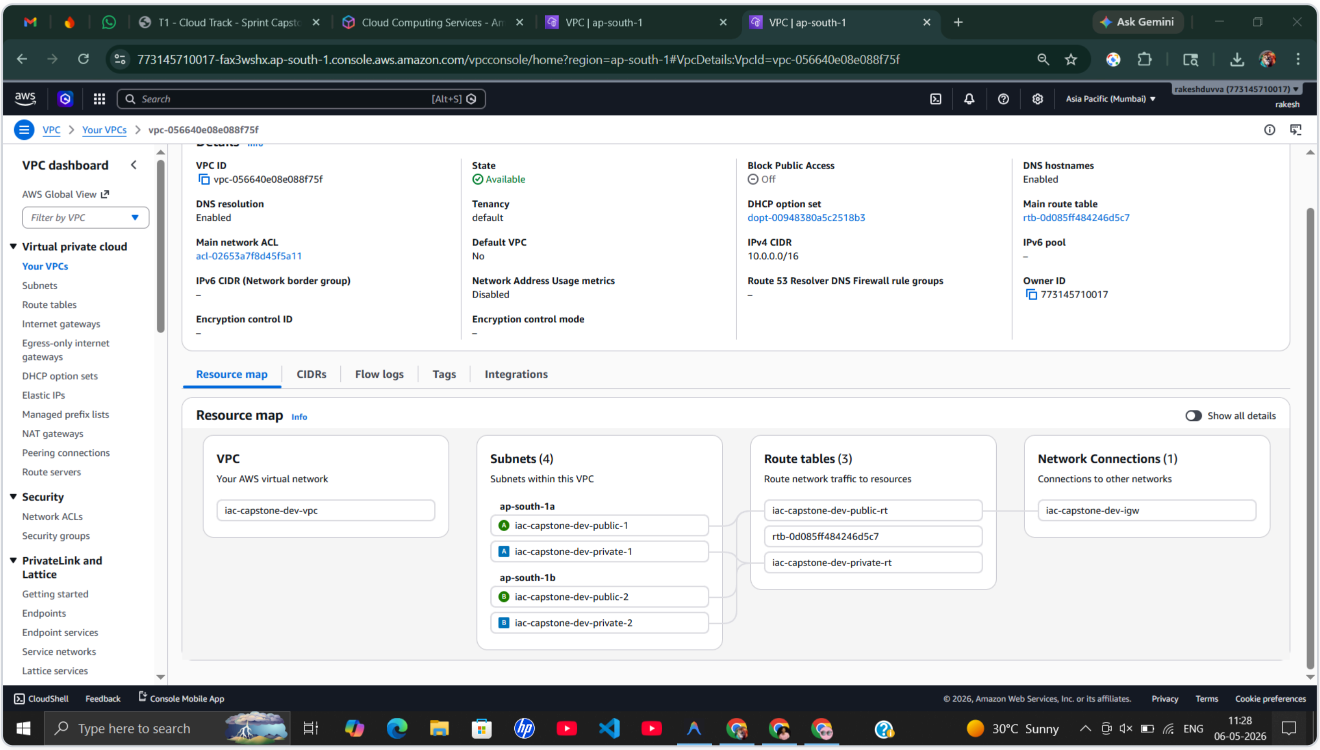

Figure 3.5: *Full-page zoomed-out view of VPC resource map — all 4 subnets, 3 route tables, and IGW visible in a hierarchical layout.*

Subnets

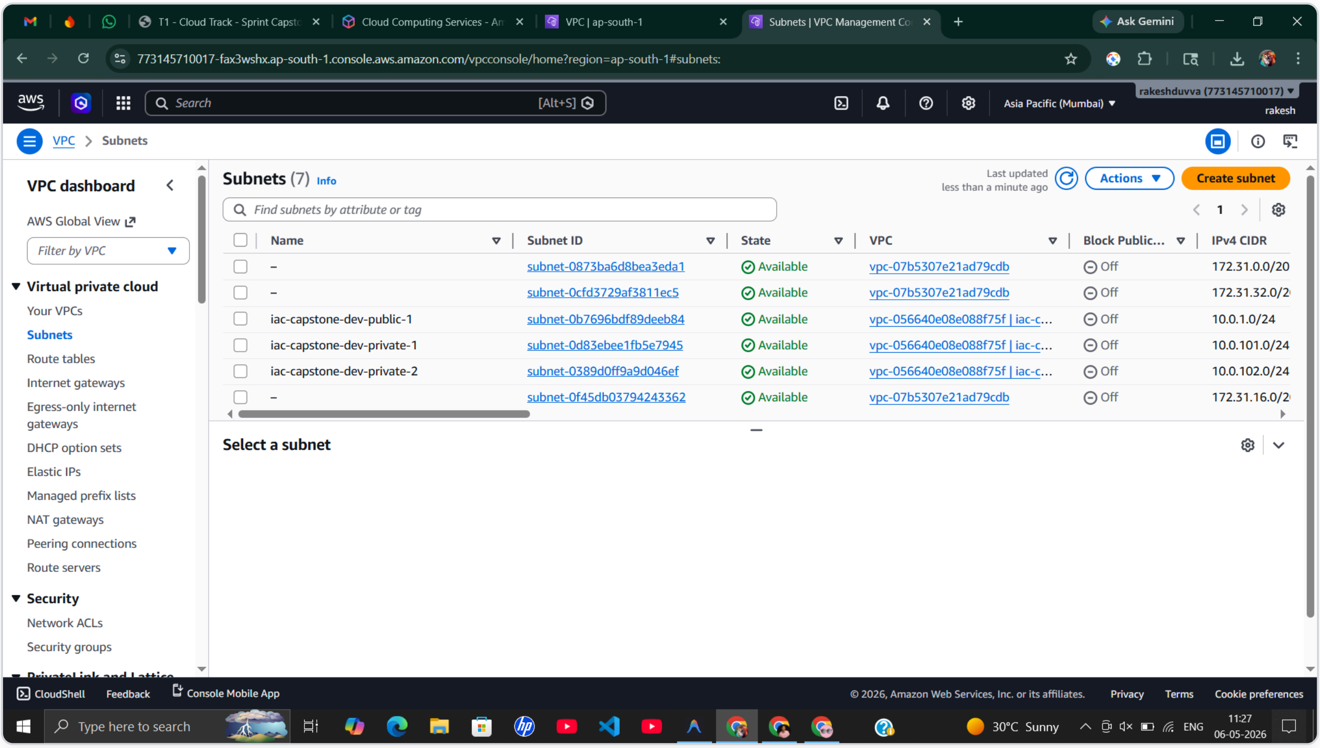

Figure 3.6: *Subnets console showing all project subnets (public-1, private-1, private-2) in "Available" state within the VPC.*

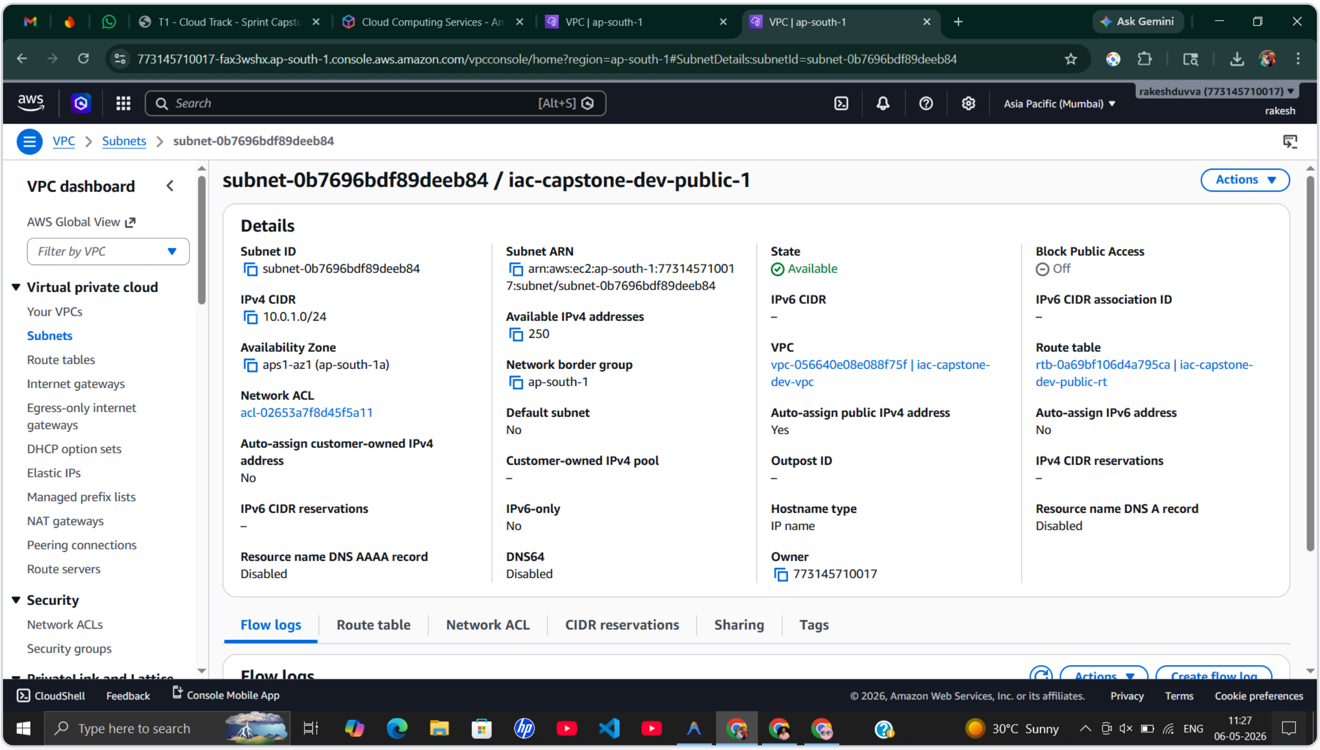

Figure 3.7: *Public Subnet 1 — CIDR:* `10.0.1.0/24` *, AZ: ap-south-1a, auto-assign public IPv4: Yes, 250 available IPs.*

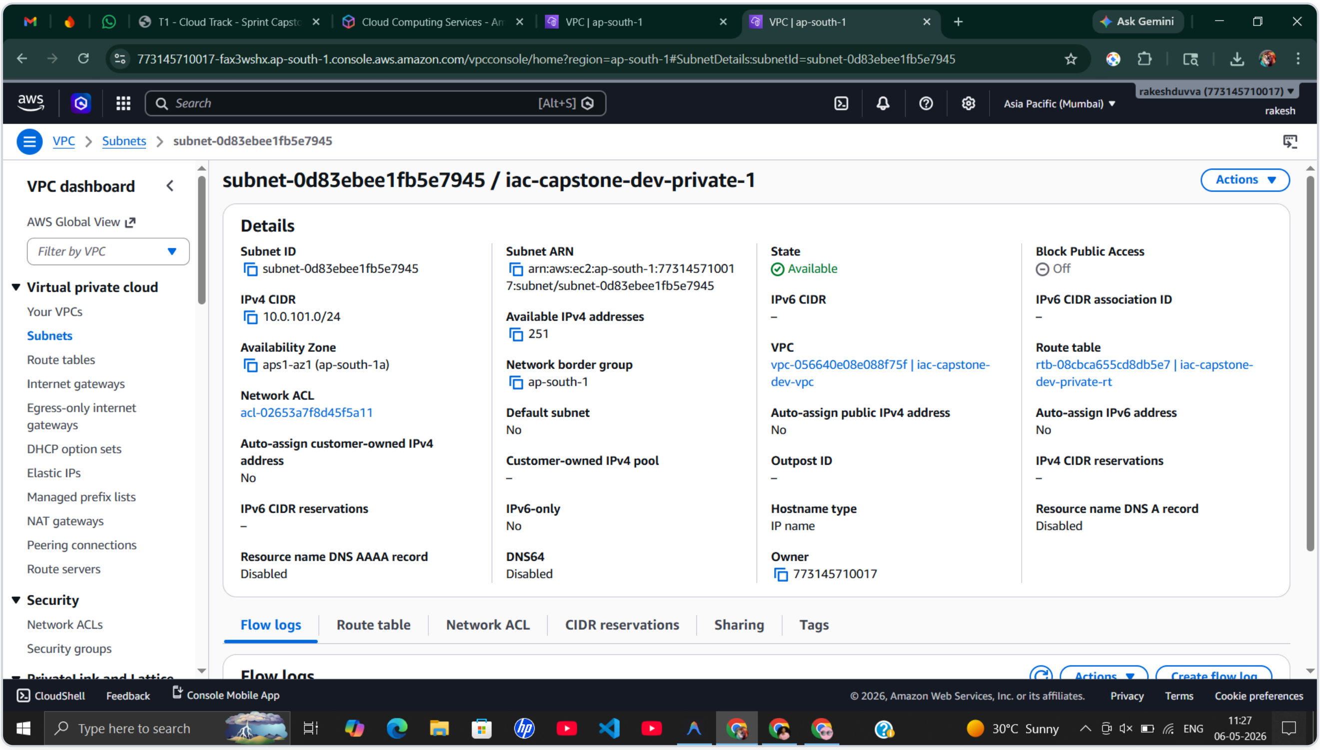

Figure 3.8: *Private Subnet 1 — CIDR:* `10.0.101.0/24` *, AZ: ap-south-1a, auto-assign public IPv4: No, 251 available IPs.*

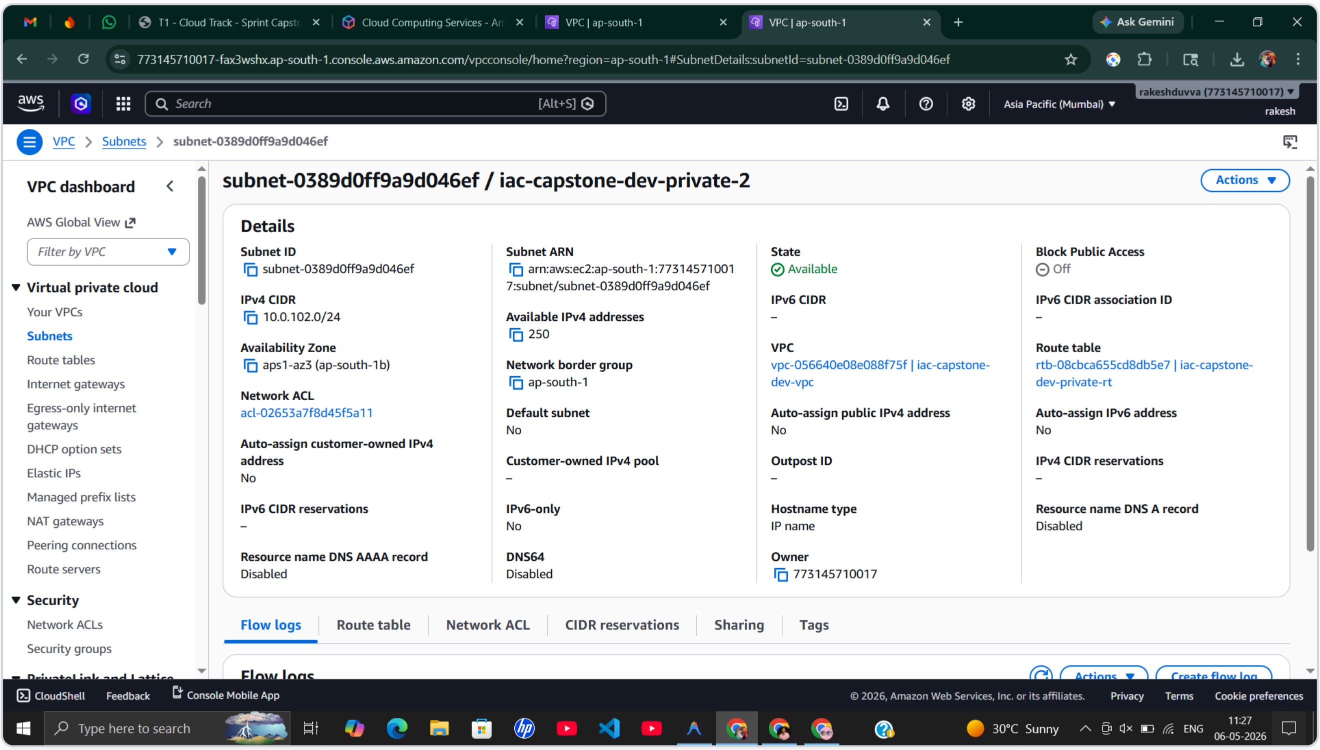

Figure 3.9: *Private Subnet 2 — CIDR:* `10.0.102.0/24` *, AZ: ap-south-1b, auto-assign public IPv4: No, 250 available IPs.*

3.2 Compute

EC2 Instance

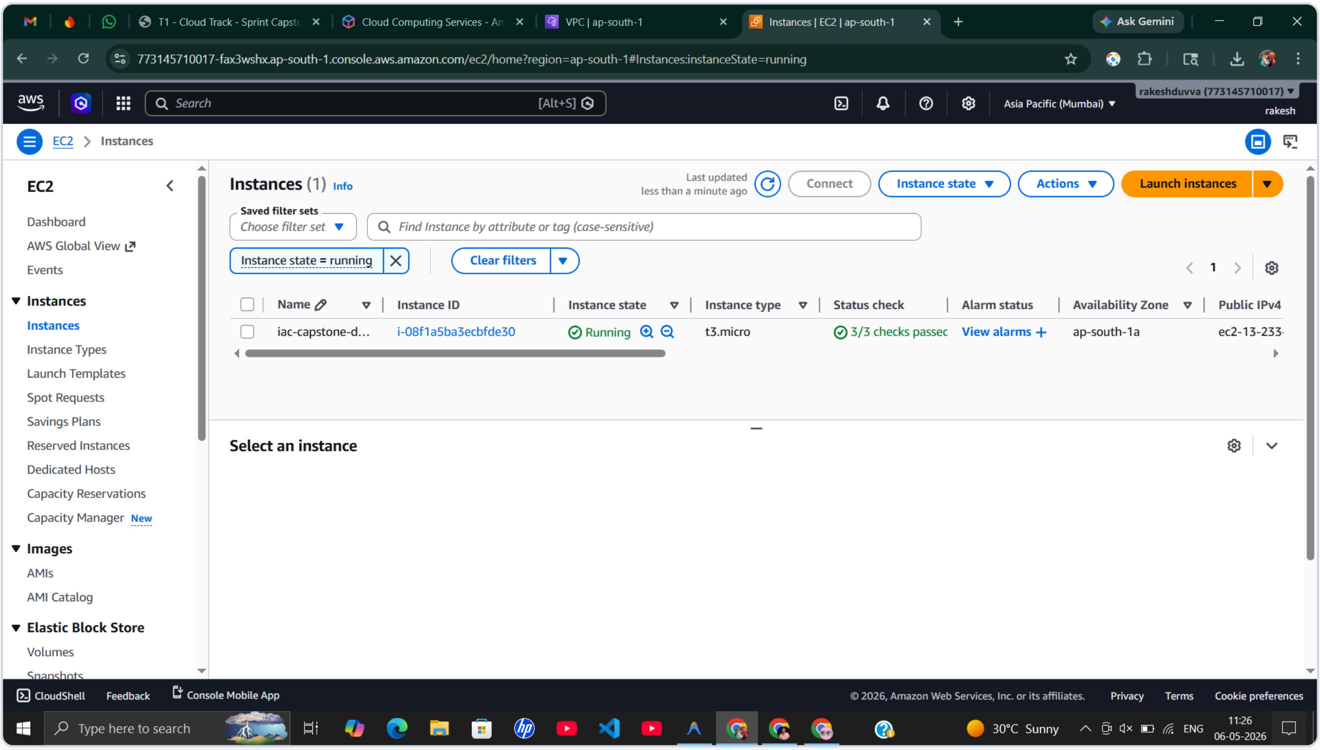

Figure 3.10: *EC2 Console showing* `iac-capstone-d...` *instance (i-08f1a5ba3ecbfde30) in "Running" state, t3.micro, ap-south-1a, 3/3 status checks passed.*

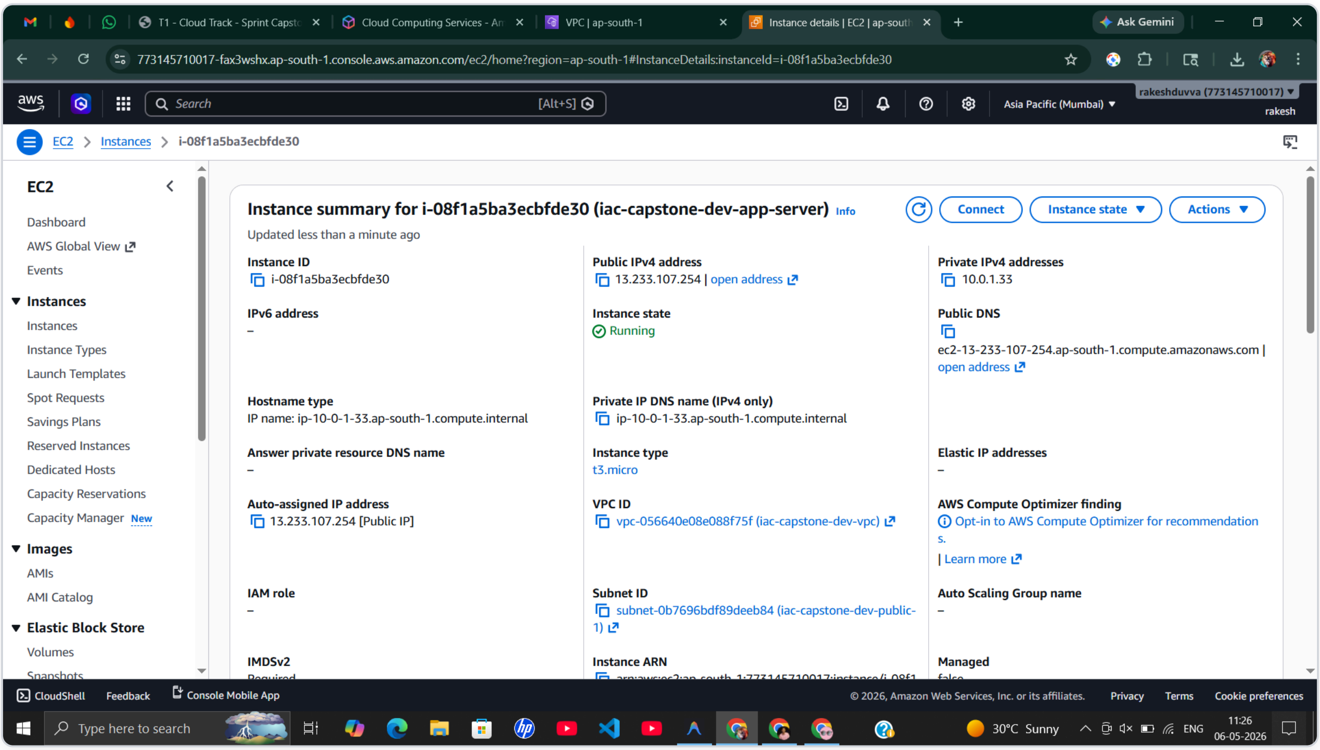

Figure 3.11: *Instance summary — Public IP:* `13.233.107.254` *, Public DNS:* `ec2-13-233-107-254.ap-south-1.compute.amazonaws.com` *, VPC: iac-capstone-dev-vpc, Subnet: iac-capstone-dev-public-1, IMDSv2: Required.*

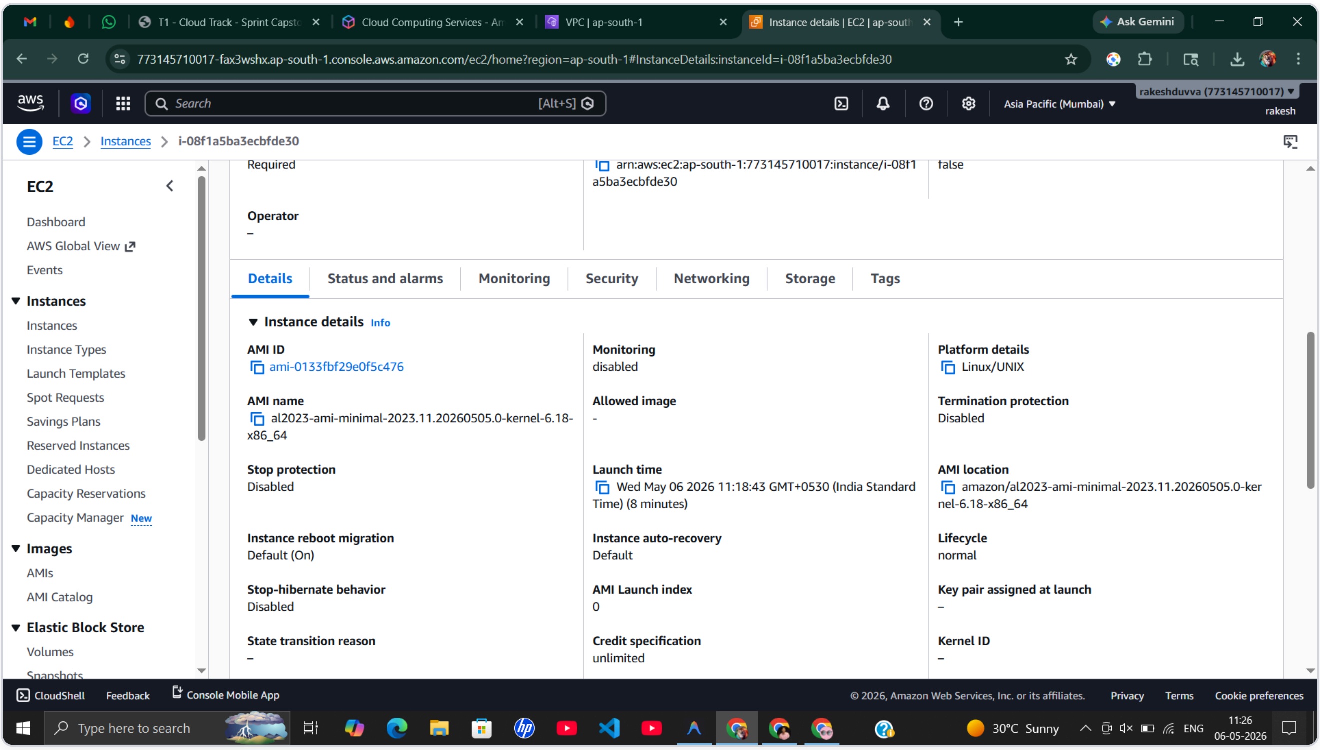

Figure 3.12: *Instance details — AMI:* `ami-0133fbf29e0f5c476` *(Amazon Linux 2023), Platform: Linux/UNIX, Launch time: May 06, 2026 11:18:43.*

Security Groups

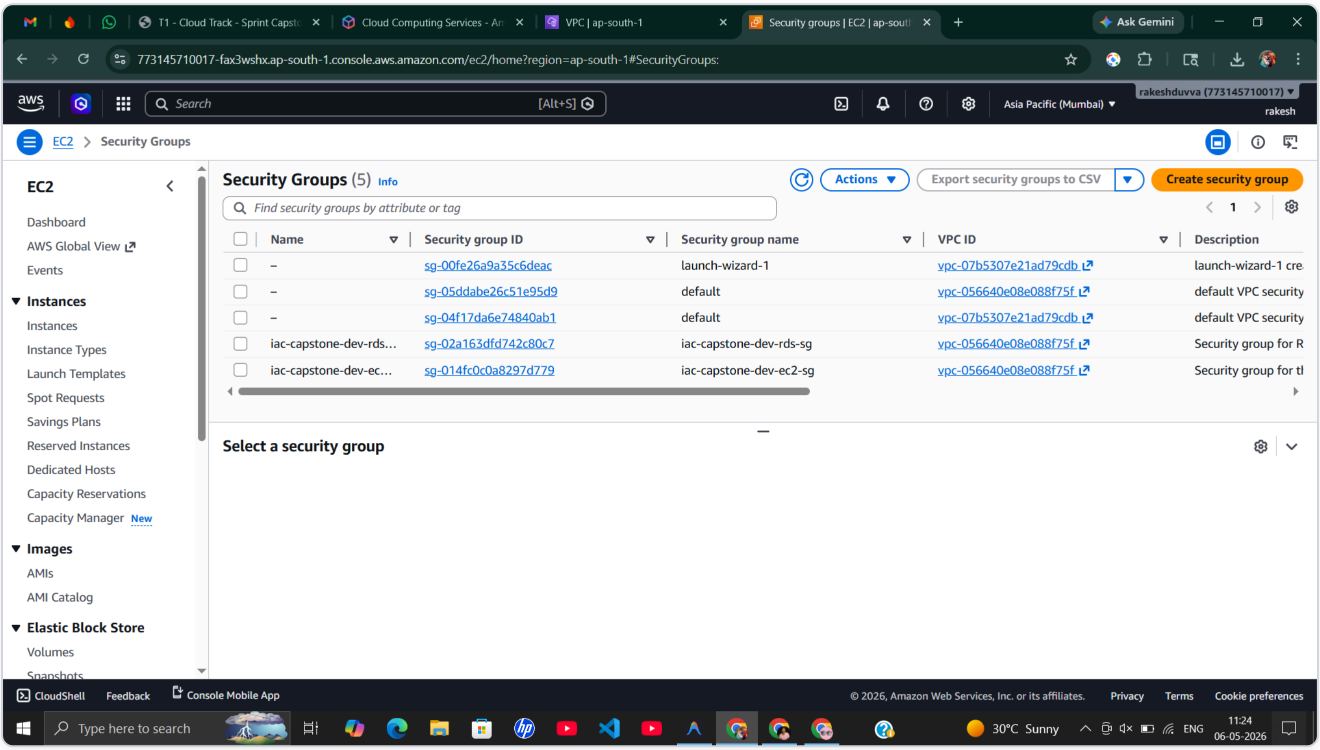

Figure 3.13: *Security Groups console listing the two Terraform-managed groups:* `iac-capstone-dev-rds-sg` *and* `iac-capstone-dev-ec2-sg` *, both attached to the project VPC.*

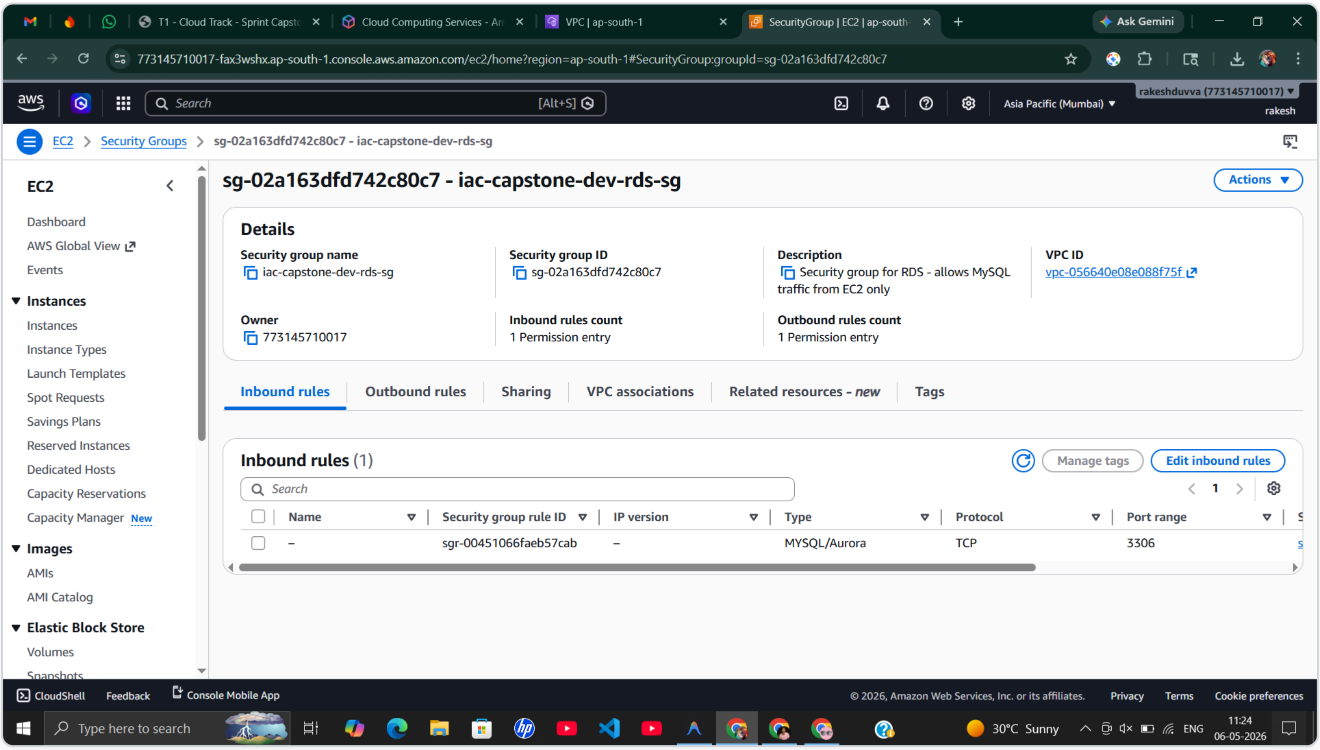

Figure 3.14: *RDS Security Group — inbound rule: MySQL/Aurora (TCP 3306) restricted to EC2 security group only (not 0.0.0.0/0). Description: "Security group for RDS - allows MySQL traffic from EC2 only".*

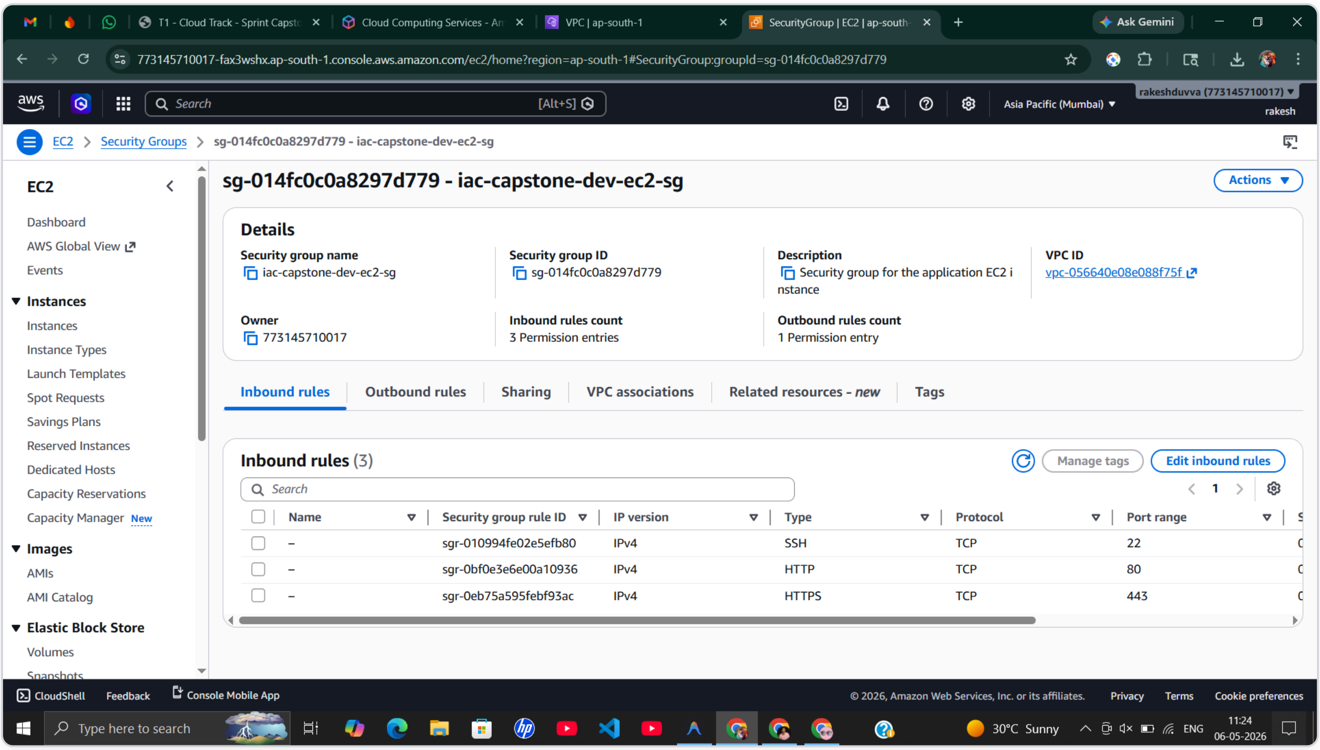

Figure 3.15: *EC2 Security Group — 3 inbound rules: SSH (TCP 22), HTTP (TCP 80), HTTPS (TCP 443). Description: "Security group for the application EC2 instance".*

3.3 Storage

S3 Bucket

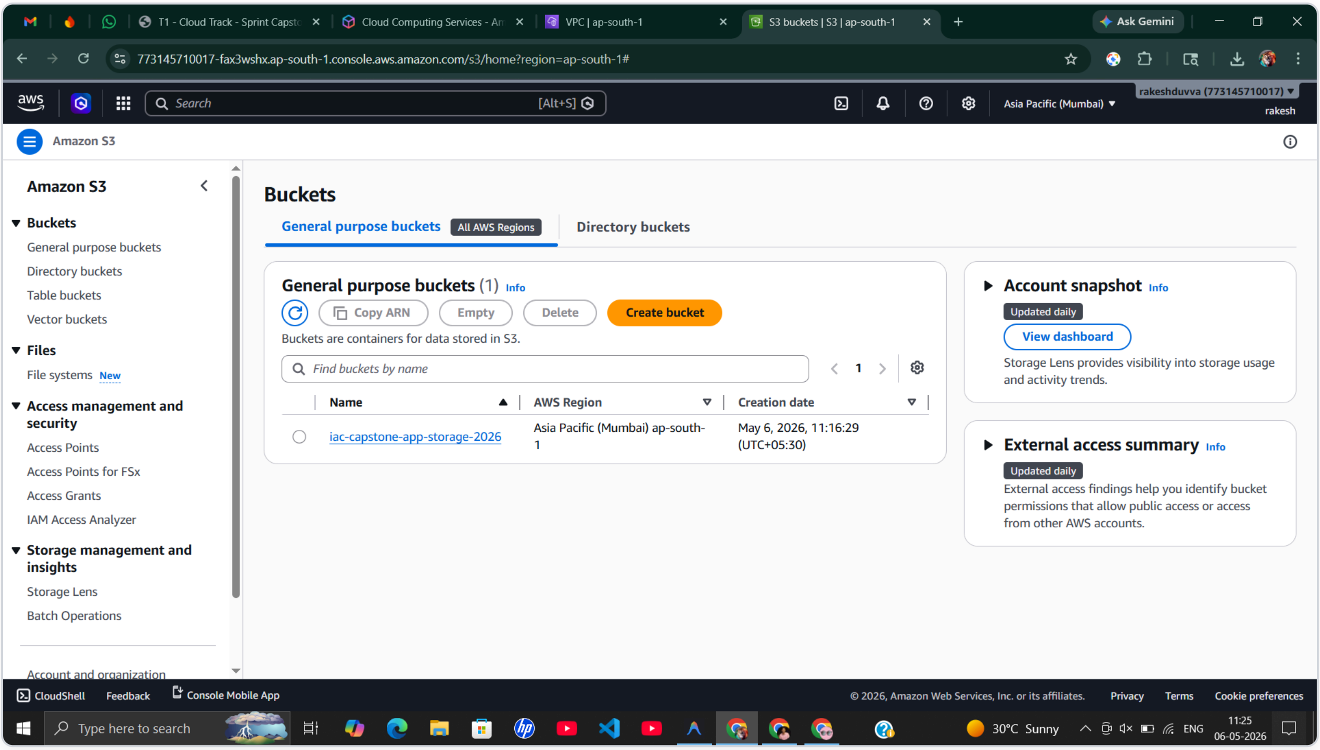

Figure 3.16: *S3 Console showing* `iac-capstone-app-storage-2026` *bucket in Asia Pacific (Mumbai) ap-south-1, created May 6, 2026.*

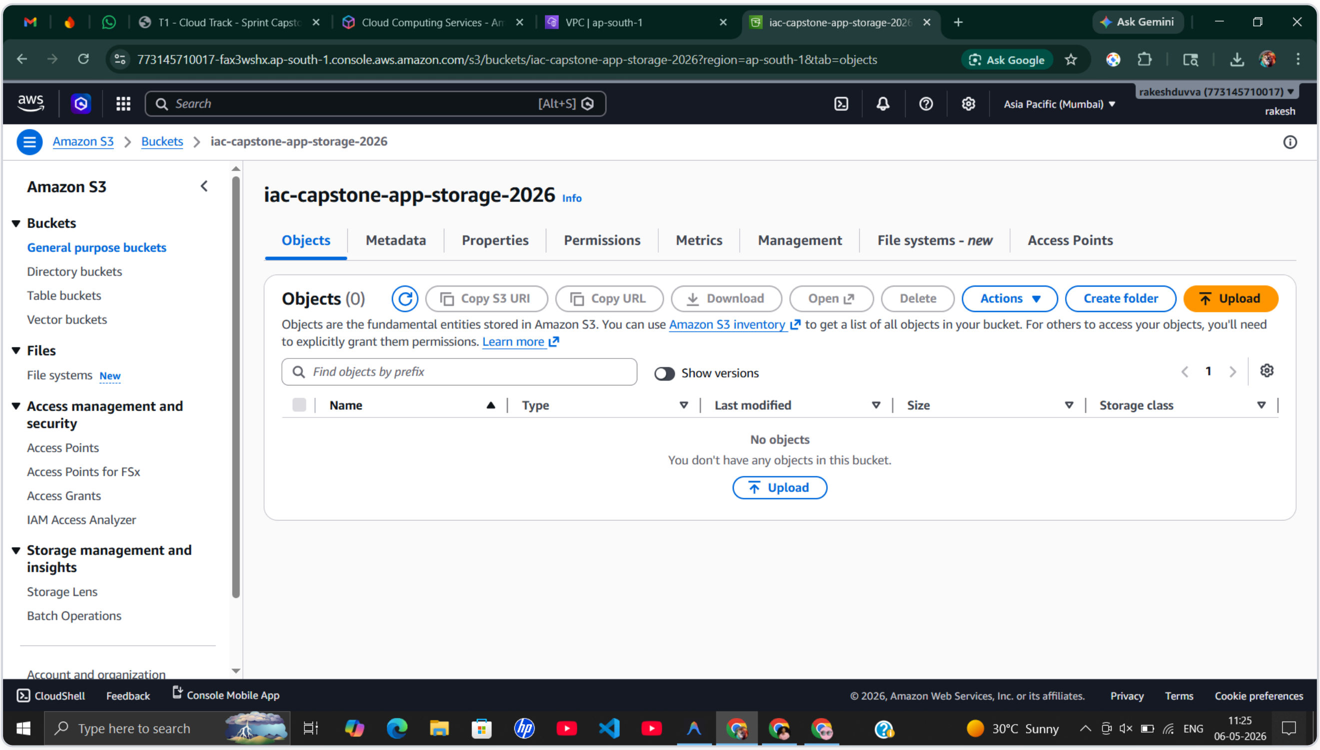

Figure 3.17: *S3 bucket Objects tab — empty bucket with versioning enabled, server-side encryption (AES-256), lifecycle rules configured, and all public access blocked.*

3.4 Database

RDS MySQL

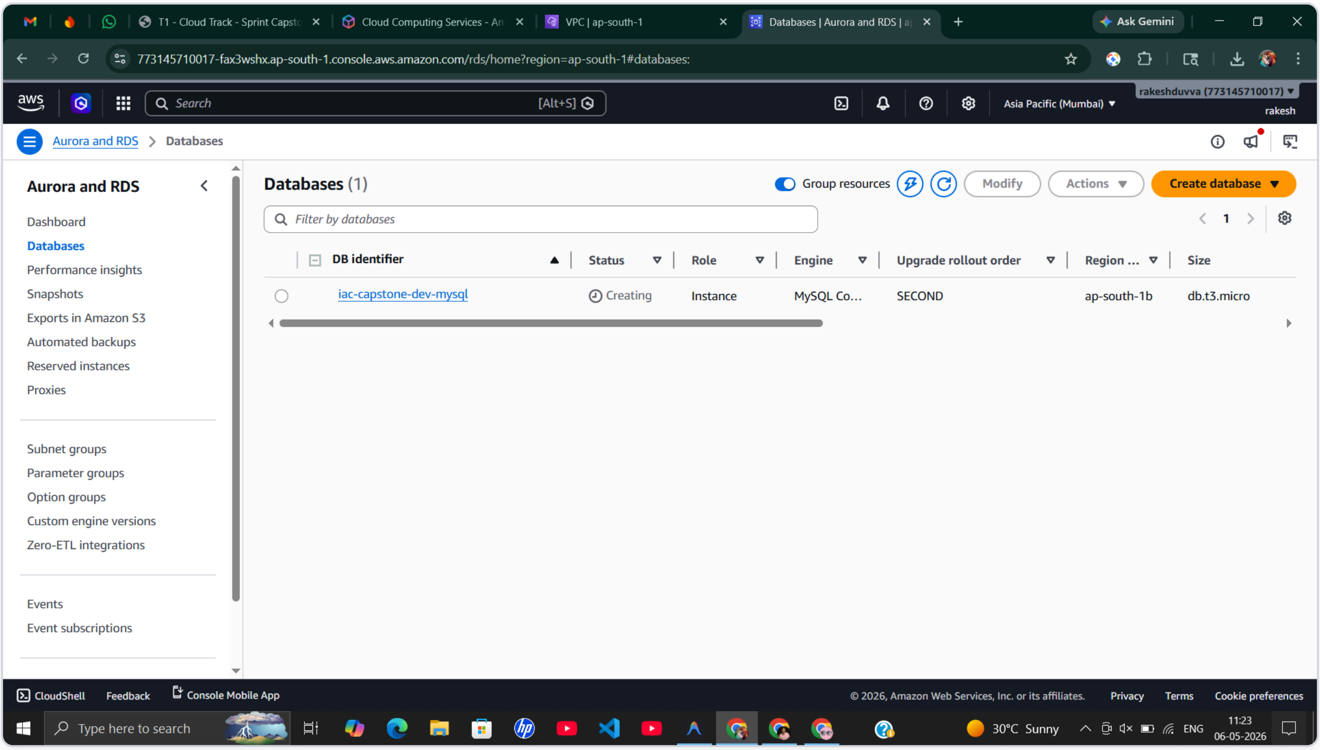

Figure 3.18: *RDS Console showing `iac-capstone-dev-mysql` in "Creating" status, MySQL Community engine, db.t3.micro, ap-south-1b.*

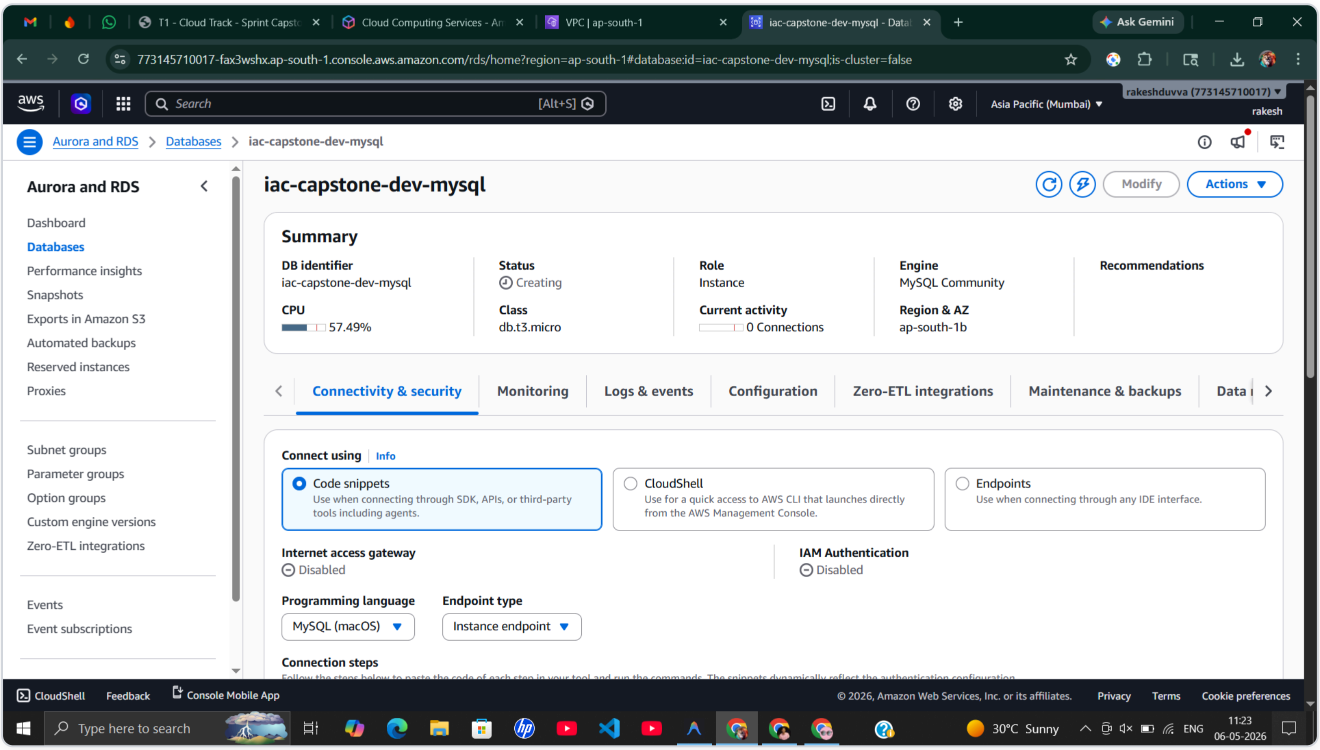

Figure 3.19: *RDS instance summary — DB identifier: iac-capstone-dev-mysql, Engine: MySQL Community, Class: db.t3.micro, Internet access: Disabled (private only).*

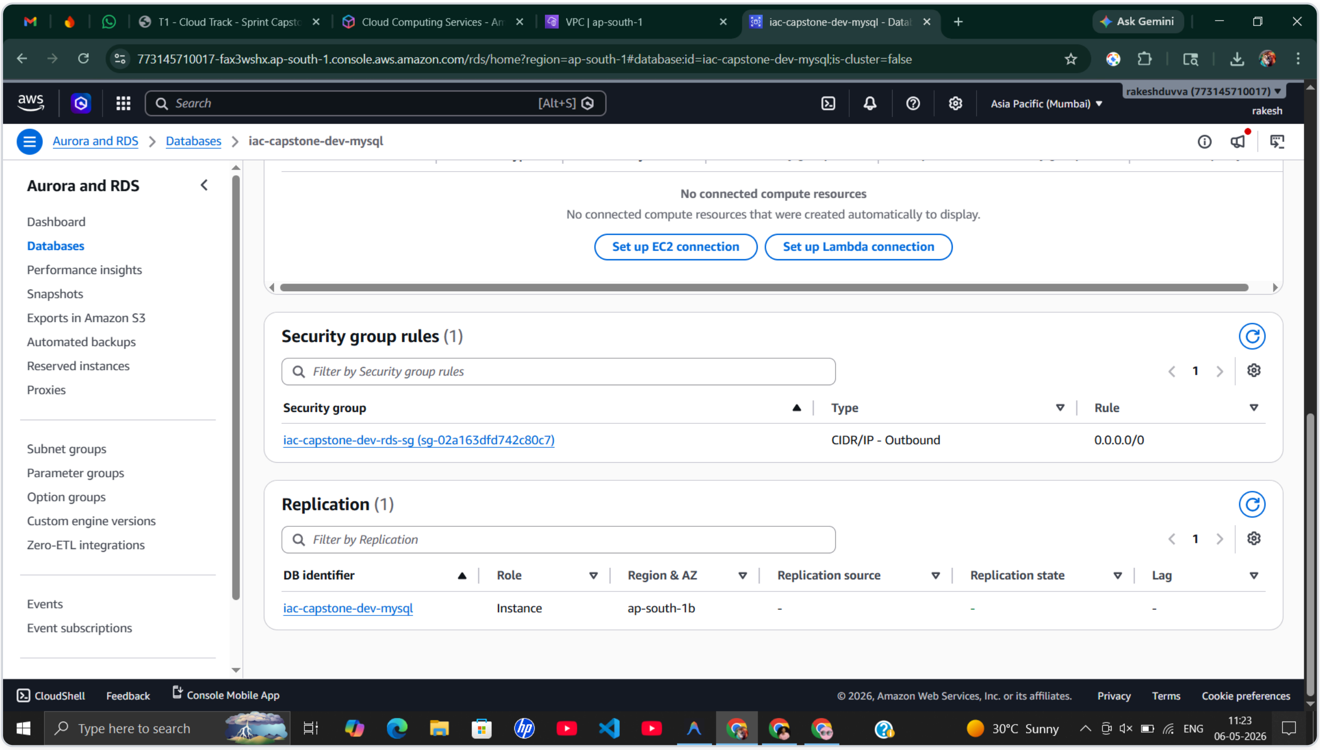

Figure 3.20: *RDS connectivity — Security group:* `iac-capstone-dev-rds-sg` *, replication role: Instance in ap-south-1b.*

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4. Infrastructure Dashboard

A custom-built web dashboard (in `web/`) that parses the `terraform.tfstate` file to visualize all 22 resources in real-time.

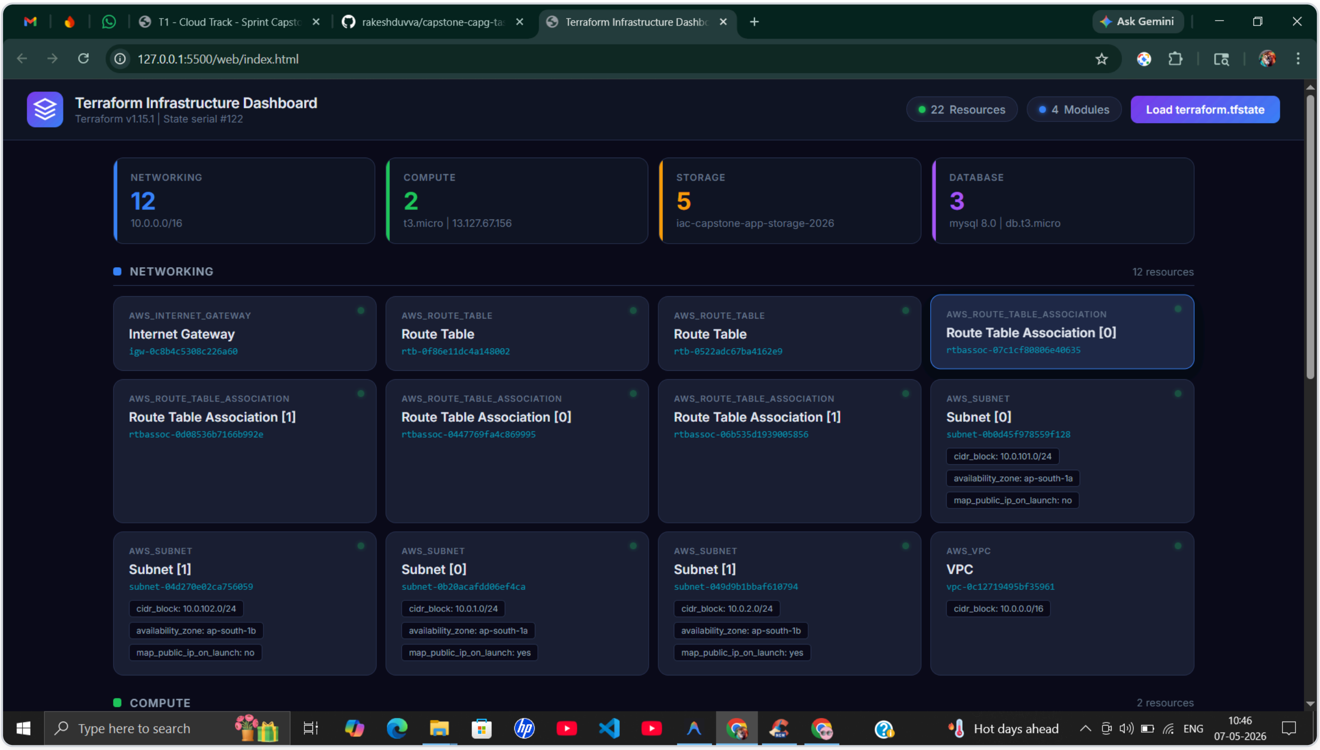

Figure 4.1: *Terraform Infrastructure Dashboard showing all 22 resources organized by module — Networking (12), Compute (2), Storage (5), Database (3). Each card displays resource type, name, ID, and key attributes.*

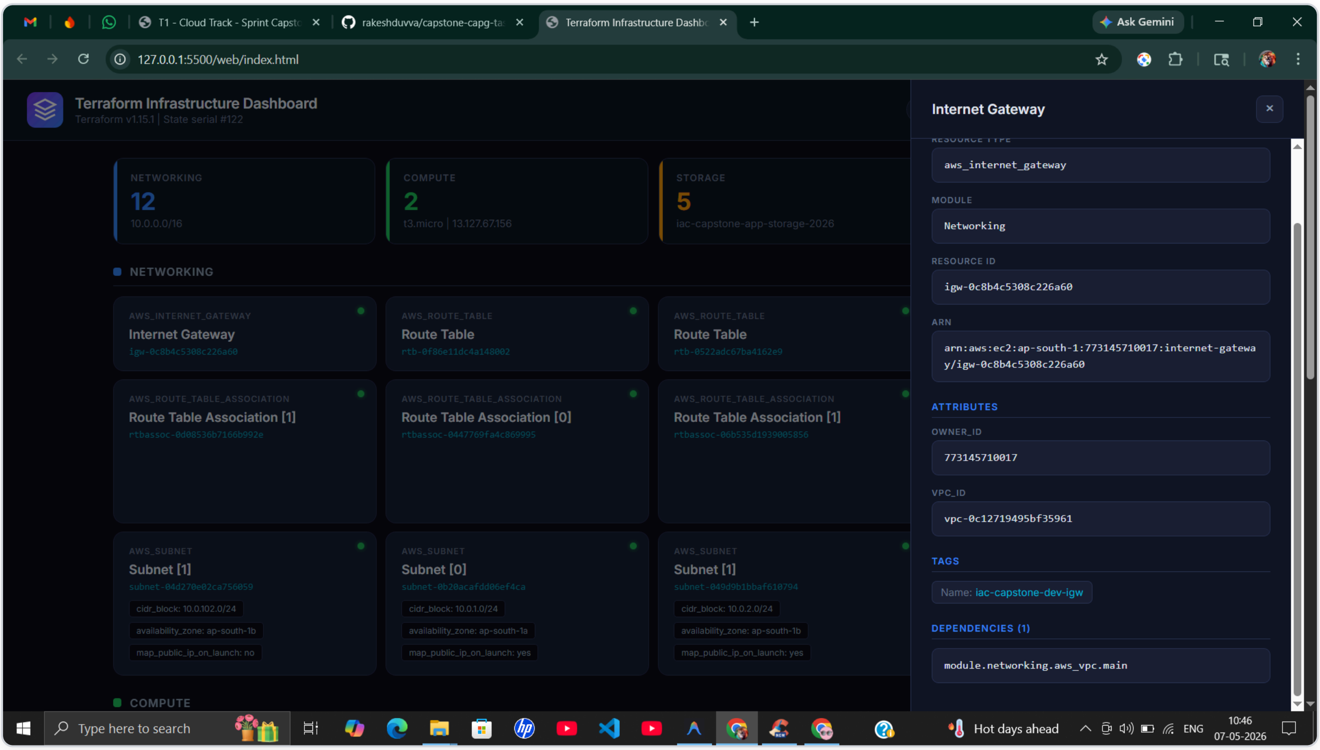

Figure 4.2: *Resource detail inspector (slide-in panel) showing Internet Gateway details: resource type, module, ID, ARN, owner, VPC ID, tags, and dependencies.*

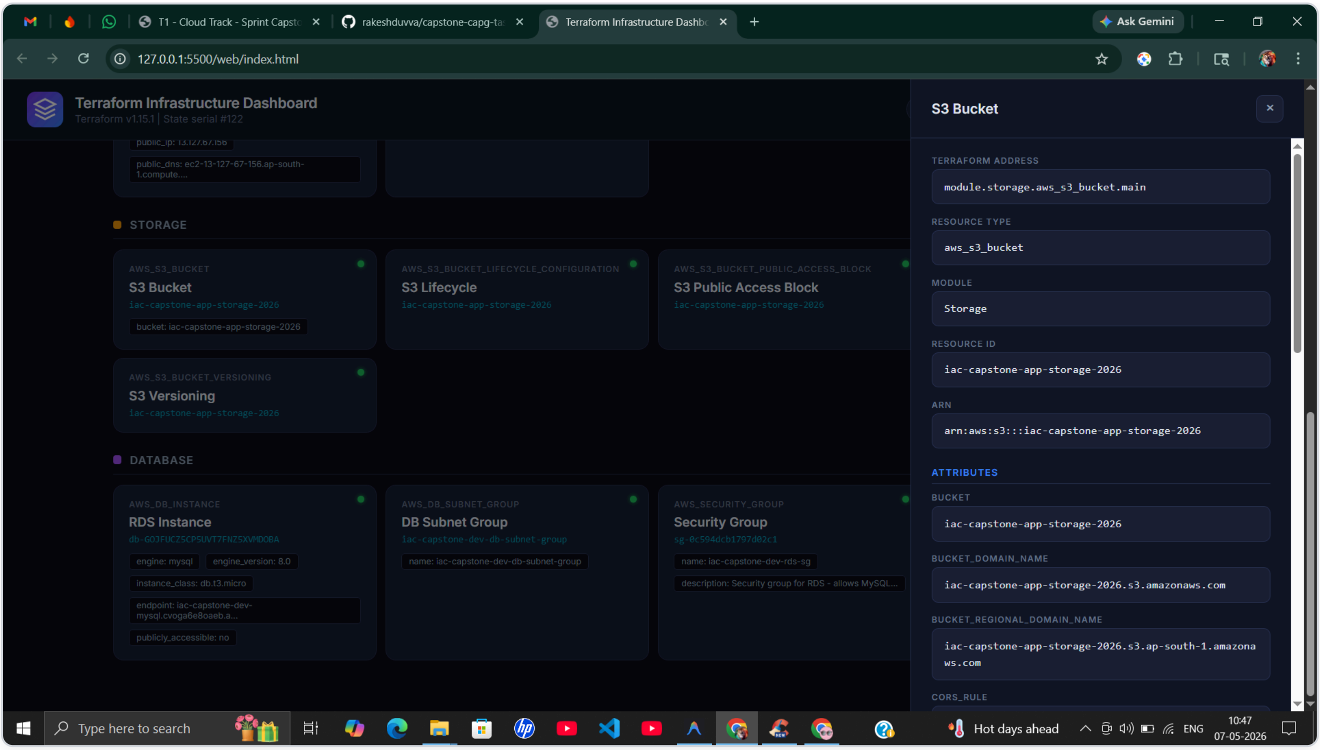

Figure 4.3: *Dashboard showing Storage and Database modules with S3 Bucket detail inspector — bucket name, domain names, regional endpoint, and CORS rules.*

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5. Summary

Aspect	Details
Modular Design	4 independent Terraform modules (networking, compute, storage, database)
Full Lifecycle	Init → Plan → Apply → Destroy demonstrated end-to-end
Security	Private subnets for RDS, restricted SGs, S3 public access blocked, encryption at rest
Policy-as-Code	7 OPA/Conftest rules enforcing security standards
Observability	Custom web dashboard for real-time infrastructure visualization
Console Verification	All resources confirmed via AWS Management Console

Resources Managed: 22 • **Modules:** 4

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